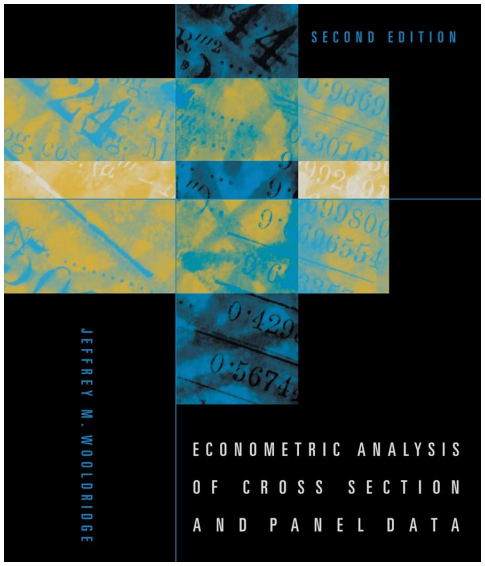
Using Panel Data Analysis to Address Important Social Questions



and now let's consider time in our analysis

Social Questions

Panel 1. Using the HILDA panel data, run a pooled regression with and without fixed effects and progressively include the variables from the dataset that you consider have an effect on the dependent variable, Health. Report the: effect(s), standard errors, sample size, level of explanation for each specification, and; interpret the results in terms of the research question.

Panel 2. Investigate whether Mental Health from past periods has an effect on current and future life satisfaction. Using the HILDA panel data, run a series of regressions of the effect of current, past and future Mental Health on current and future Life Satisfaction; include the variables from the dataset that you consider have an effect on the dependent variable. Interpret the results.

Panel 3. Using the HILDA panel data explain the U-shape in Happiness over time.

Panel Analysis

Often, we use cross-sectional data from one or a couple of periods (pooled) for analysis:

$$\boldsymbol{Health_i = \hat{\beta}_1 + \hat{\beta}_2 BMI_i + e_i}$$

$$E(\varepsilon_{it}|X_{it}) = 0 \text{ (pooled OLS assumption)}$$

The data in each cross-section often comes from different individuals. So, we cannot analyse changes at the individual (*i)* level across time.

Data are collected from the same individuals across time. Examples are the Australian HILDA and the German SOEP socio-economic panel data sets that include respondent's answers the same questions year after year.

Panel data are shaped differently from cross-sectional/data pooled from a number of periods. Panel data allow us to do analysis of changing effects over time (*t*).

Task: What is a Balanced panel?
When should you use a balanced panel?

Pooled Regression

	Age		Age + Age^2		Usual Suspects		Usual Suspects + Health		Kitchen Sink	
Variable:	coefficient	t-value	coefficient	t-value	coefficient	t-value	coefficient	t-value	coefficient	t-value
age	0.0107	33.37	-0.0320	-18.98	-0.0554	-29.57	-0.0403	-22.72	-0.0311	-16
age*age			0.0004	25.79	0.0007	33.95	0.0006	30.26	0.0005	23.98
income					0.0805	15.00	0.0329	6.03	0.0309	5.69
male					-0.1471	-13.52	-0.1016	-9.93	-0.0981	-9.53
education					-0.0232	-7.52	-0.0582	-19.74	-0.0606	-20.6
number of children					-0.0590	-10.81	-0.0623	-12.15	-0.0600	-11.25
married					0.4496	36.20	0.3624	30.32	0.2514	14.85
employed					0.1355	9.27	-0.1338	-9.2	-0.1166	-8.00
unemployed					-0.3244	-9.36	-0.4208	-12.92	-0.3560	-10.87
regional income							0.0000	5.87	0.0000	5.2
home owner							0.1397	10.95	0.1253	9.82
imputed rent							-0.0001	-0.42	-0.0001	-0.43
health							-0.5278	-88.46	-0.5251	-88.37
invalid							-0.1073	-7.92	-0.1046	-7.76
family death									0.0110	0.68
divorced									-0.0771	-3.51
separated									-0.2947	-9.12
partner dead									0.0237	0.78
just married									0.1175	3.7
just divorced									-0.1853	-2.72
just separated									-0.4209	-15.2
spouse just died									-0.2597	-4.59
just had a baby									0.1278	3.81
pregnant									0.1382	4.88
just fired from job									-0.2875	-9.52
constant	7.4240	477.31	8.3255	217.8	8.0870	110	10.0957	135.44	10.0082	132.16
R^2	0.0151		0.0241		0.0531		0.1661		0.1745	
Adjusted R^2			0.0240		0.0530		0.1660		0.1742	

Table B4: The determinants of Life Satisfaction; Pooled OLS regression results for all individuals in the HILDA; *N = 75,529*

Pooled Regression

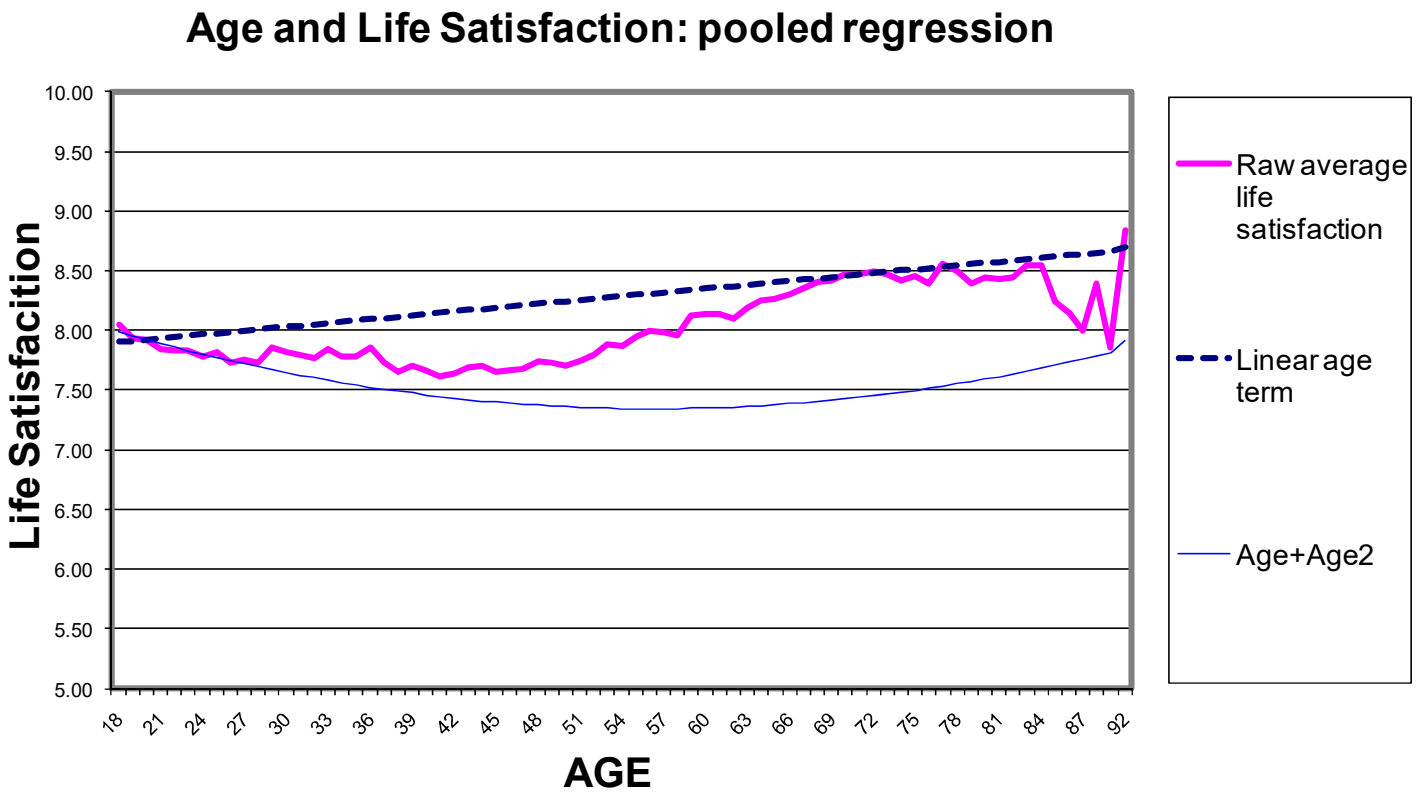
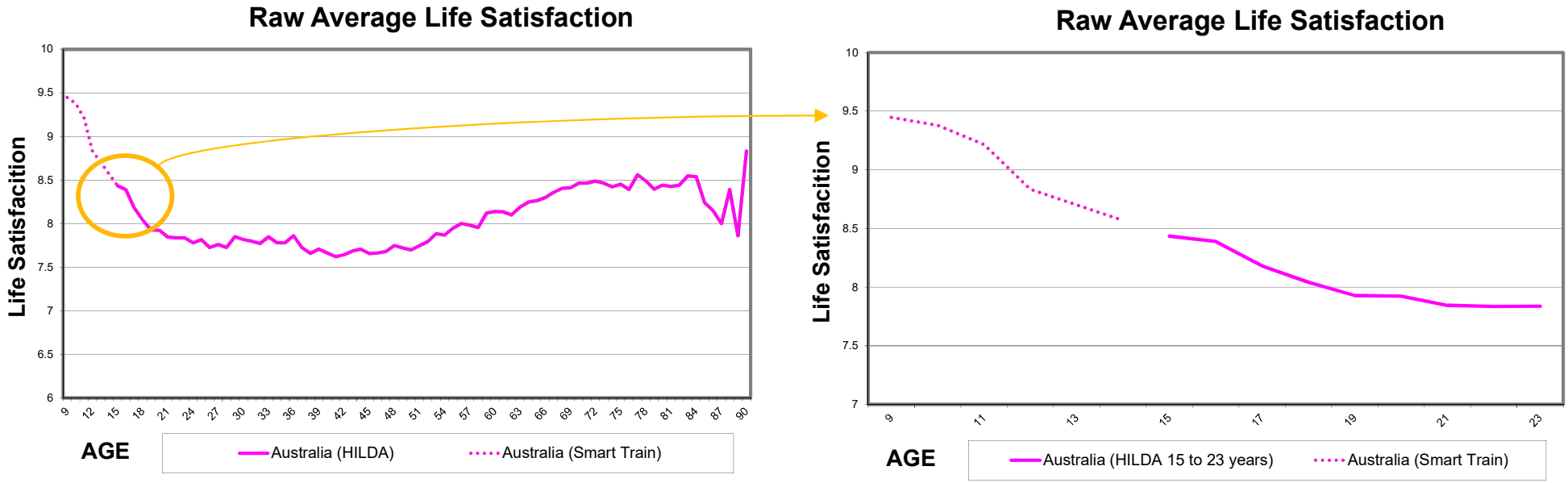


Figure 1b: Life satisfaction in the HILDA for the pooled sample

Task: What causes this U-shape in happiness over time?

Pooled Regression

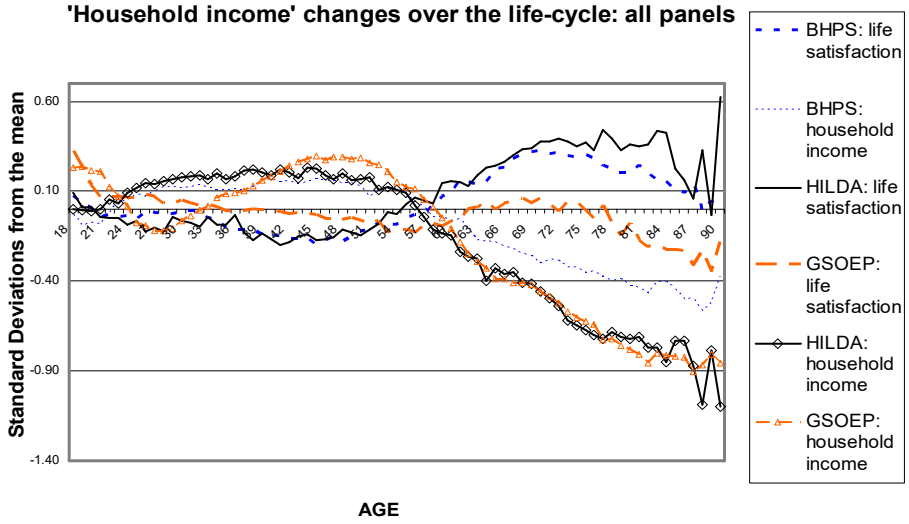


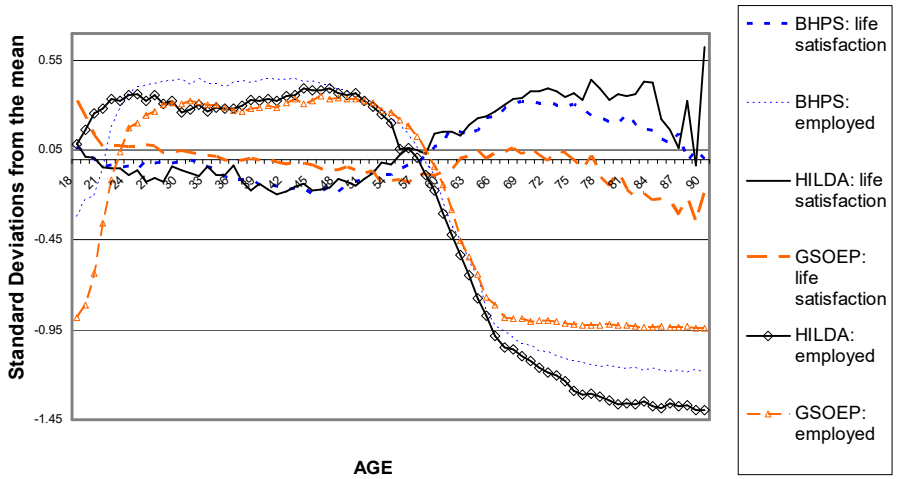
Figures 2(a) & (b): Average Life Satisfaction for 9 to14 year-old Australian children in the 'Smart Train' data and 15 to 93 year-old Australians in the 2002-2013 HILDA panel data

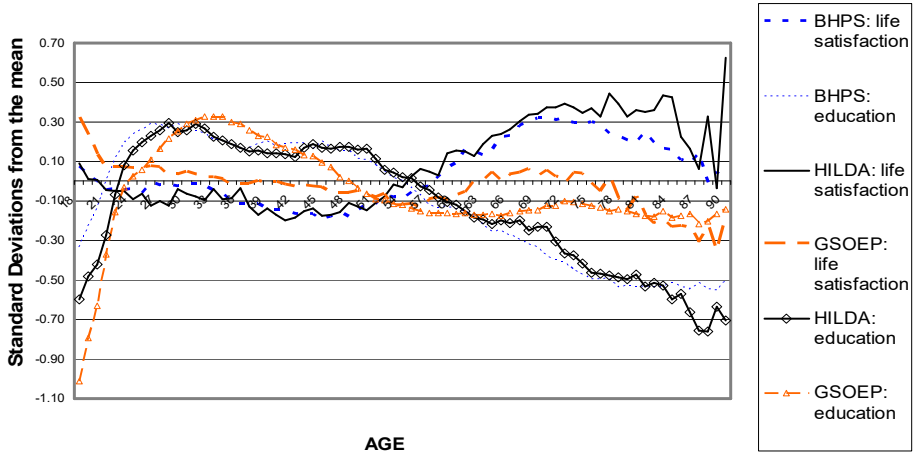
Task: What causes this U-shape in happiness over time?

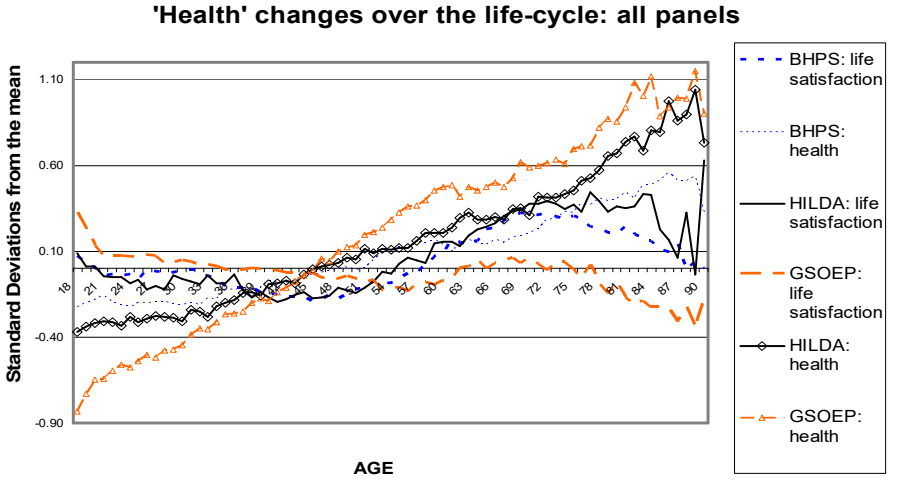
Income?

Pooled Regression









Task: What causes this U-shape in happiness over time?

Panel Analysis (Time)

Wide format:

country	gdp2020	gdp2021	gdp2022
USA	21000	21080	21100
China	19000	20000	21000
Vietnam	900	1009	1100

Long format is required for panel analysis in Stata:

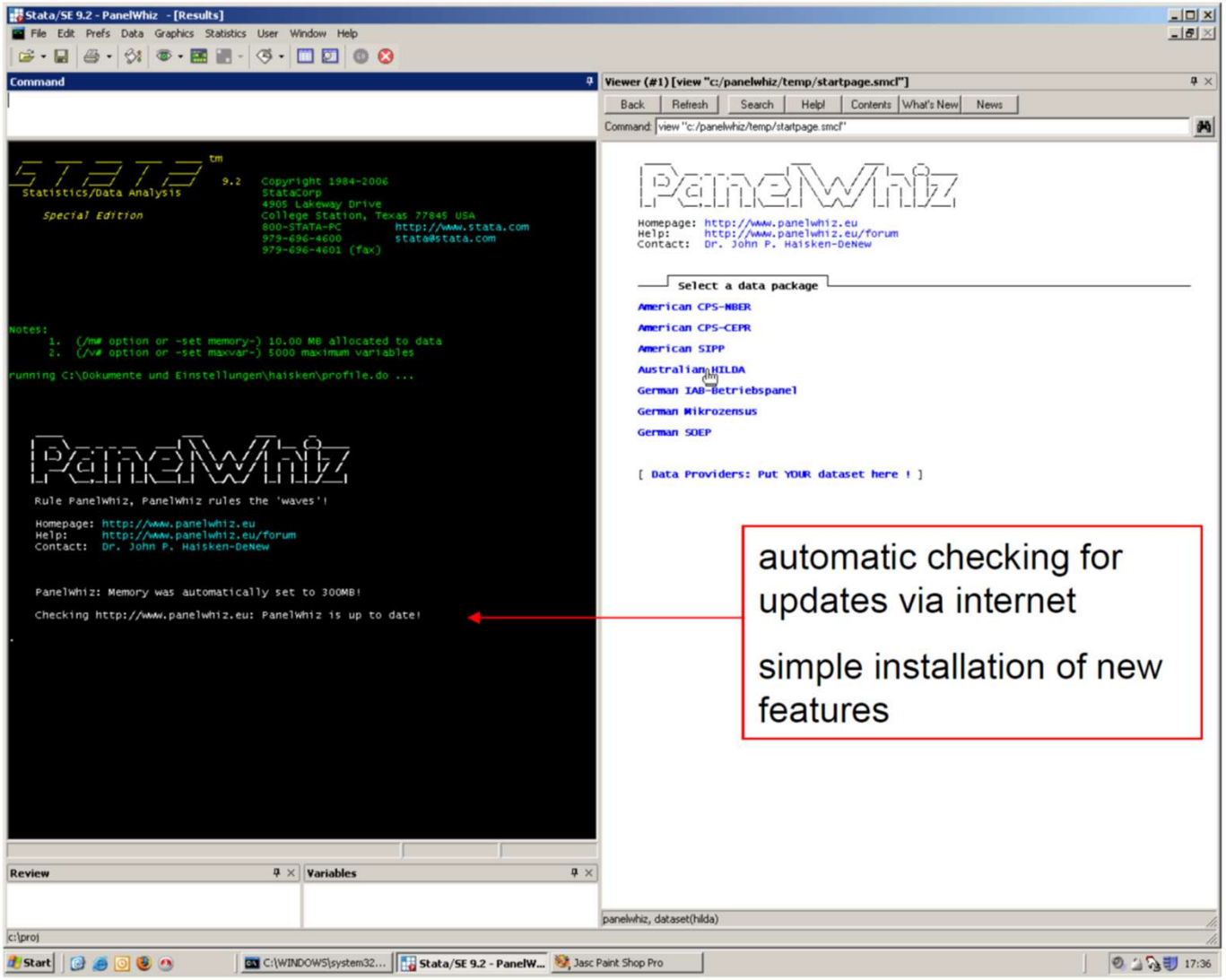
persnr	year	lifesat	mental_healtl
100003	2006	8	80
100003	2008	7	64
100003	2009	[10] Totally satisfied	76
100003	2011	7	80
100003	2012	8	72
100003	2014	7	60
100003	2015	8	64
100003	2018	7	76
100003	2019	7	84
100003	2020	8	72

Stata: help reshape

reshape long lifesat mental_health, i(persnr) j(year) // to go from wide to long

reshape wide lifesat mental_health, i(persnr) j(year) // to go from long to wide

PanelWhiz


Stata/SE 9.2 - PanelWhiz - [Results]
File Edit Prefs Data Graphics Statistics User Window Help
Command
Viewer (#1) [view "c:/panelwhiz/temp/startpage.smcl"]
Back Refresh Search Help! Contents What's New News
Command: view "c:/panelwhiz/temp/startpage.smcl"
Statistics/Data Analysis
Special Edition
9.2 Copyright 1984-2006
StataCorp
4905 Lakeway Drive
College Station, Texas 77845 USA
800-STATA-PC http://www.stata.com
979-696-4600 stata@stata.com
979-696-4601 (fax)
Notes:
1. (/m# option or -set memory-) 10.00 MB allocated to data
2. (/v# option or -set maxvar-) 5000 maximum variables
running C:\Dokumente und Einstellungen\haisken\profile.do ...
PanelWhiz
Rule PanelWhiz, PanelWhiz rules the 'waves'!
Homepage: http://www.panelwhiz.eu
Help: http://www.panelwhiz.eu/forum
Contact: Dr. John P. Haisken-DeNew
PanelWhiz: Memory was automatically set to 300MB!
Checking http://www.panelwhiz.eu: PanelWhiz is up to date!
Homepage: http://www.panelwhiz.eu
Help: http://www.panelwhiz.eu/forum
Contact: Dr. John P. Haisken-DeNew
Select a data package
American CPS-NBER
American CPS-CEPR
American SIPP
Australian HILDA
German IAB-Betriebspanel
German Mikrozensus
German SOEP
[Data Providers: Put YOUR dataset here !]
automatic checking for updates via internet
simple installation of new features
Review
Variables
panelwhiz, dataset(hilda)
c:\proj
Start
C:\WINDOWS\system32...
Stata/SE 9.2 - PanelW...
Jasc Paint Shop Pro
17:36

Panel Analysis – Pooled Regression

- Put all years into a pooled dataset and run an OLS; ignore time.
- Run one year at a time (cross-sectional analysis.

$$\boldsymbol{LS_i = \beta_1 + \beta_2 X_i + u_i + e_i}$$

$$E(\varepsilon_{it}|X_{it}) = 0\,(pooled\ OLS\ assumption)$$

Pooled OLS ignores individual heterogeneity; regression results will be biased, if individual-specific traits affect the outcome.

Panel Analysis – Fixed Effects (within)

Panel data allow us to get an idea of how things change over time (*t*). Life Satisfaction example:

$$LS_{it} = \beta_1 + \beta_2 X_{it} + u_i + e_{it}$$

where multiple covariates, (*x)* variables (e.g. income, female) as well as fixed (*u*), or random, effects can be applied to a model to explain Life Satisfaction (LS) of the average individual across time.

Fixed Effects (u) are those information in a variable that do not change from one period to the next: ie: LS_{it} - LS_{it-1} ; often called a 'first' difference:

$$LS_{it} - LS_{it-1} = \beta_1 + \beta_2 X_{it} + u_i + v_{it} \; ; t=1 \ldots, T$$

$$E(\varepsilon_{it} | u_i, X_{it}) = 0 \text{ (FE assumption)}$$

With Fixed effects (within) model the explanatory variables (x) are not permitted to correlate with time invariant effects (u) or the error term (v).

Stata: xtset persnr year
xtreg lifesat lnhousehold_income female, fe cluster(persnr)

Panel Analysis: Fixed Effects

Stata: xtset persnr year
local cluster persnr
local IV6 age lnhousehold_income female educyrs bachelor mental_health
eststo fe1_`d':  xtreg lifesat `IV6' , fe cluster(`cluster') // (5)
esttab, margin compress b(%9.4f) se(%9.4f) star r2 (%9.4f) nogaps
parentheses star(+ 0.10 * 0.05 ** 0.01) *

	(1) Life Satisfaction	(2) Life Satisfaction	(3) Life Satisfaction	(4) Life Satisfaction	(5) Life Satisfaction
age	-0.0006	-0.0006	-0.0018*	-0.0018*	0.0027**
	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0007)
Ln Household Income	0.0371**	0.0371**	0.0372**	0.0372**	0.0303**
	(0.0043)	(0.0043)	(0.0043)	(0.0043)	(0.0039)
female		0.0000	0.0000	0.0000	0.0000
		(.)	(.)	(.)	(.)
educ years			-0.0085**	-0.0088**	-0.0163**
			(0.0015)	(0.0026)	(0.0024)
bachelor				0.0027	0.0327+
				(0.0188)	(0.0173)
Mental Health					0.0258**
					(0.0003)
N	230857	230857	230857	230857	230857
R-squared	0.0008	0.0008	0.0009	0.0009	0.0911

Marginal effects; Standard errors in parentheses
+ p<0.10, * p<0.05, ** p<0.01

Fixed Effect Regression

	Age		Age + Age^2		Usual Suspects		Usual Suspects + Health		Kitchen Sink	
Variable:	coefficient	t-value	coefficient	t-value	coefficient	t-value	coefficient	t-value	coefficient	t-value
age	-0.0167	-6.90	-0.0076	-1.02	-0.0173	-2.12	-0.0202	-2.50	-0.0033	-0.40
age*age			-0.0001	-1.30	-0.00001	-0.17	-0.00007	0.94	-0.00006	-0.72
income					0.0257	4.26	0.0191	3.22	0.0151	2.55
education					-0.0204	-1.72	-0.0198	-1.71	-0.0246	-2.11
number of children					-0.0329	-2.24	-0.0312	-2.18	-0.0436	-2.92
married					0.2056	6.76	0.1963	6.52	-0.1095	-2.84
employed					0.0536	2.33	0.0238	1.06	0.0394	1.75
unemployed					-0.1344	-3.19	-0.1422	-3.43	-0.1208	-2.91
regional income							0.0000	3.30	0.0000	2.65
home owner							0.0687	3.01	0.0640	2.83
imputed rent							-0.0001	-0.79	-0.0001	-0.62
health							-0.2694	-28.16	-0.2679	-28.13
invalid							-0.0492	-3.29	-0.0488	-3.28
family death									-0.0046	-0.33
divorce									-0.1608	-2.71
separated									-0.3479	-4.83
partner dead									-0.3512	-3.85
just married									0.1361	4.56
just divorced									-0.2707	-2.94
just separated									-0.3063	-8.45
spouse just died									-0.2326	-3.02
just had a baby									0.1338	4.79
pregnant									0.1258	4.94
just fired from job									-0.0389	-1.22
constant	8.6701	78.75	8.4827	47.34	8.5967	41.04	9.2482	44.12	9.1148	43.45
Overall R^2	0.0151		0.0175		0.0047		0.0344		0.0325	

Table B6: The determinants of Life Satisfaction for Australians in the HILDA; Fixed-effect Regressions – entire sample, *N* = 75,529

Panel Analysis – Random Effects (within & between)

$$LS_{it} = \beta_1 + \beta_2 X_{it} + \alpha_i + u_{it} \; ; t=1 \ldots, T$$

Random Effects has a much stronger assumption, the explanatory variables (x), random effects (∝) are assumed to not be correlated with the error term (ε):

$$E(\varepsilon_{it} \mid \propto_i, X_{it}) = E(\varepsilon_{it} \mid X_{it}) \; ; E(\varepsilon_{it} \mid X_{it}) = 0 \; (assumes \; pooled \; OLS \; assumption)$$

Random Effects (RE) Must be completely uncorrelated with independent variables; results will be biased if unobserved traits correlate with predictors.

Stata: xtset persnr year
xtreg lifesat lnhousehold_incomemental_health, re **cluster(female)**

Panel Analysis: Random Effects

Stata: xtset persnr year
local cluster persnr
local IV6 age lnhousehold_income female educyrs bachelor mental_health
eststo fe1_`d':  xtreg lifesat `IV6' , re cluster(`cluster') // (5)
esttab, margin compress b(%9.4f) se(%9.4f) star r2 (%9.4f) nogaps parentheses star(+ 0.10 * 0.05 ** 0.01) *

	(1)	(2)	(3)	(4)	(5)
	Life Satisfaction	**Life Satisfaction**	**Life Satisfaction**	**Life Satisfaction**	**Life Satisfaction**
age	0.0011**	0.0011**	0.0007*	0.0007*	-0.0003
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0003)
Ln Household Income	0.0491**	0.0492**	0.0488**	0.0488**	0.0419**
	(0.0041)	(0.0041)	(0.0041)	(0.0041)	(0.0037)
female		0.0222	0.0223	0.0222	0.1079**
		(0.0140)	(0.0140)	(0.0140)	(0.0117)
educ years			-0.0081**	-0.0085**	-0.0282**
			(0.0014)	(0.0023)	(0.0021)
bachelor				0.0037	0.0538**
				(0.0160)	(0.0143)
Mental Health					0.0291**
					(0.0003)
N	230857	230857	230857	230857	230857

Marginal effects; Standard errors in parentheses
+ p<0.10, * p<0.05, ** p<0.01

Panel Analysis: Pooled OLS; XTREG, re/fe

	(5) OLS	(5) re	(5) fe
	Life Satisfaction	**Life Satisfaction**	**Life Satisfaction**
age	0.0001	-0.0003	0.0027**
	(0.0003)	(0.0003)	(0.0007)
Ln Household Income	0.0712**	0.0419**	0.0303**
	(0.0050)	(0.0037)	(0.0039)
female	0.1325**	0.1079**	0.0000
	(0.0116)	(0.0117)	(.)
educ years	-0.0492**	-0.0282**	-0.0163**
	(0.0031)	(0.0021)	(0.0024)
bachelor	0.0703**	0.0538**	0.0327+
	(0.0191)	(0.0143)	(0.0173)
Mental Health	0.0390**	0.0291**	0.0258**
	(0.0004)	(0.0003)	(0.0003)
N	230857	230857	230857
R-squared	0.2418		0.0911

Marginal effects; Standard errors in parentheses
+ p<0.10, * p<0.05, ** p<0.01

Task: Which Test: Pooled OLS; XTREG, re/fe?

Pooled OLS vs. Fixed Effects (F-Test): Run a standard F-test to see if individual intercepts are equal. If the test is significant, Pooled OLS is rejected in favor of Fixed Effects.

regress price mpg weight
estat hettest, fstat

Pooled OLS vs. Random Effects (Breusch-Pagan LM Test): Test whether the variance of the individual-specific error components is zero. If it is not zero, Pooled OLS is rejected in favor of Random Effects.

xtreg DV IV, fe
xttests // Breusch-Pagan statistic for cross-sectional independence in the residuals of a fixed effect regression model.
Breusch-Pagan LM test for random effects. Null hypothesis: No random effects (pooled OLS is adequate). If **p-value < 0.05**, random effects are preferred.

Fixed Effects vs. Random Effects (Hausman Test):
Hausman Test evaluates whether the individual errors are correlated with the predictors. Significant p-value ($p < 0.05$): Rejects the RE assumption. Use Fixed Effects. Insignificant p-value ($p > 0.05$): Fails to reject. Use Random Effects.
xtreg DV IV, re
hausman fixed ., sigmamore // Test the appropriateness of the random-effects estimator

Task: What causes this U-shape in happiness over time?

Figure 1b: Life satisfaction in the HILDA for the pooled sample



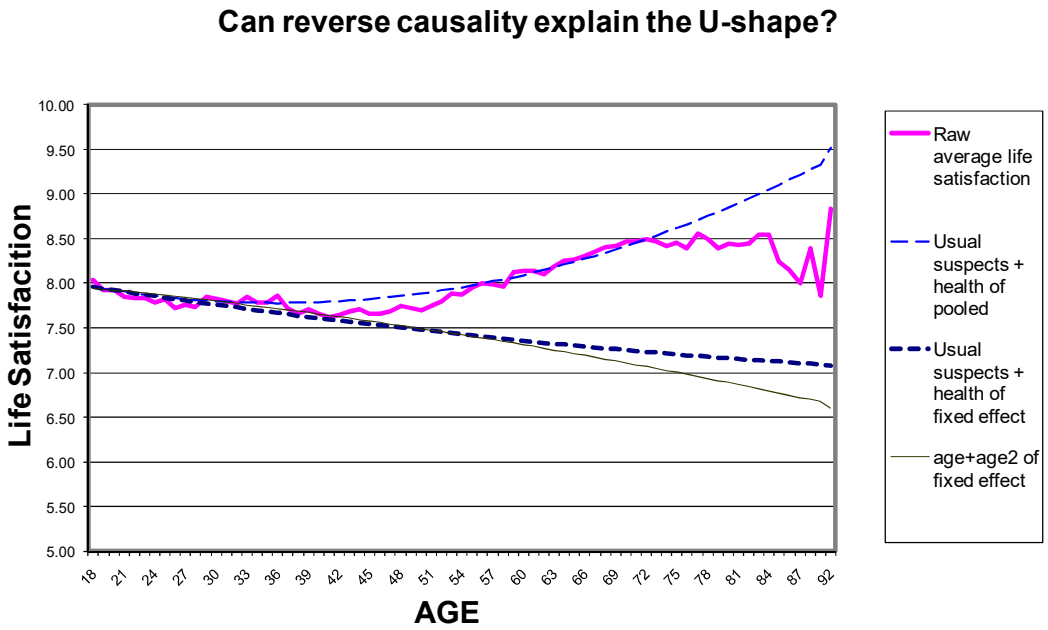


Figure 4b: Life satisfaction in the HILDA for the balanced panel including fixed-effects

Task: What causes this U-shape in happiness over time?

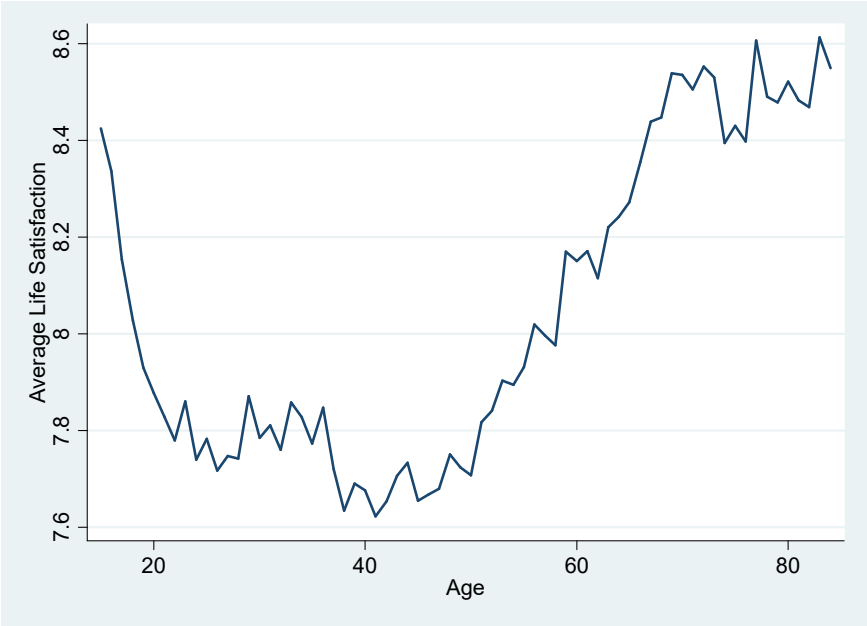


Figure 1a: Average happiness for Australians aged 15 to 84

Life Satisfaction and Stress

$$Stress_t = \sum_s SRRS_s * Le_{st-1}$$

Stress Level	Life Event
.87	***Death of a spouse***
.79	***Death of a close family member***
.78	***Major injury or illness to self***
.76	***Detention in gaol or other institution***
.72	***Major injury or illness to close family member***
.71	Foreclosure on a loan/mortgage
.71	***Divorce***
.70	***Victim of crime***
.69	Victim of police brutality
.69	Infidelity
.69	***Experiencing domestic violence/sexual abuse***
.66	***Separation with spouse/mate***
.66	***Reconciliation with spouse/mate***
.64	***Being fired/laid-off/unemployed***
.62	***Experiencing financial problems/difficulties***
.61	***Death of a close friend***

$$GS_t = C + \delta(Stress_t) + \varepsilon_t$$

Task: What causes this U-shape in happiness over time?

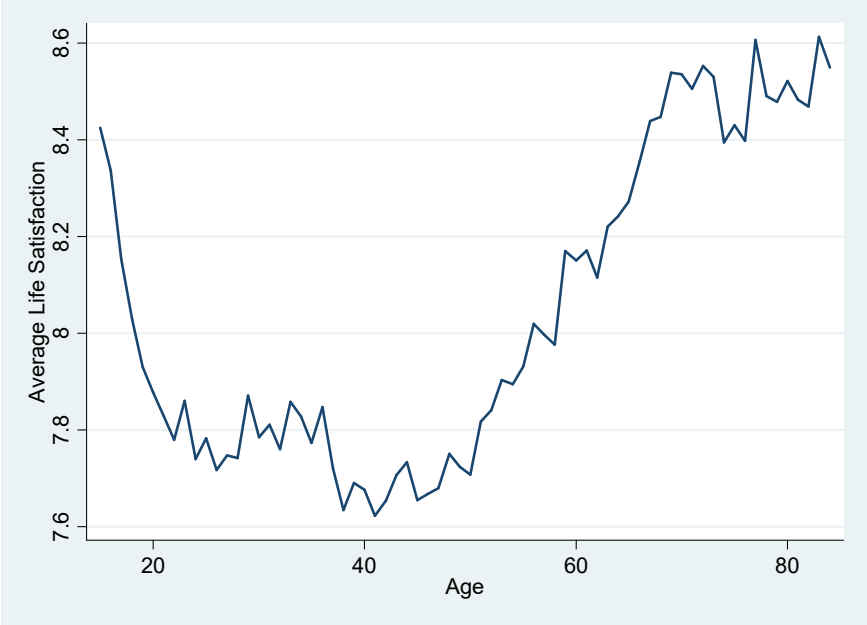


Figure 1a: Average happiness for Australians aged 15 to 84

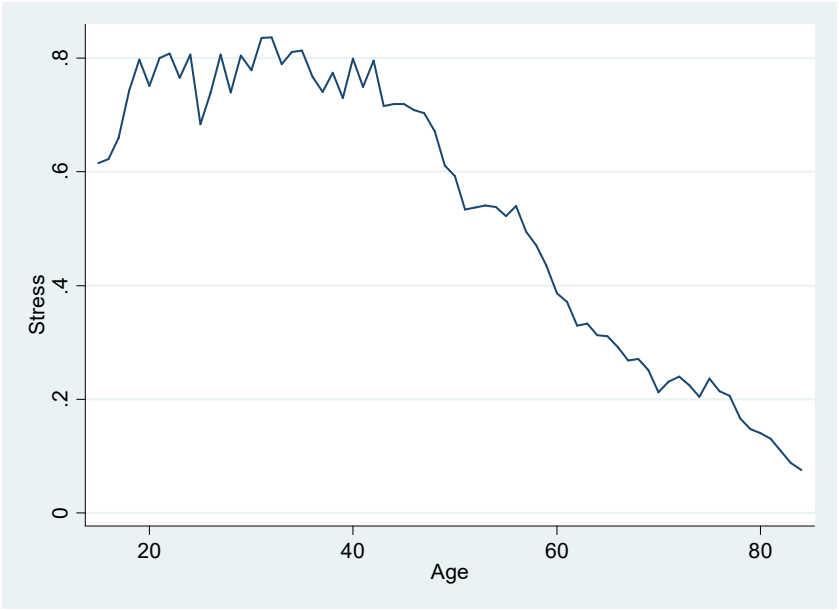
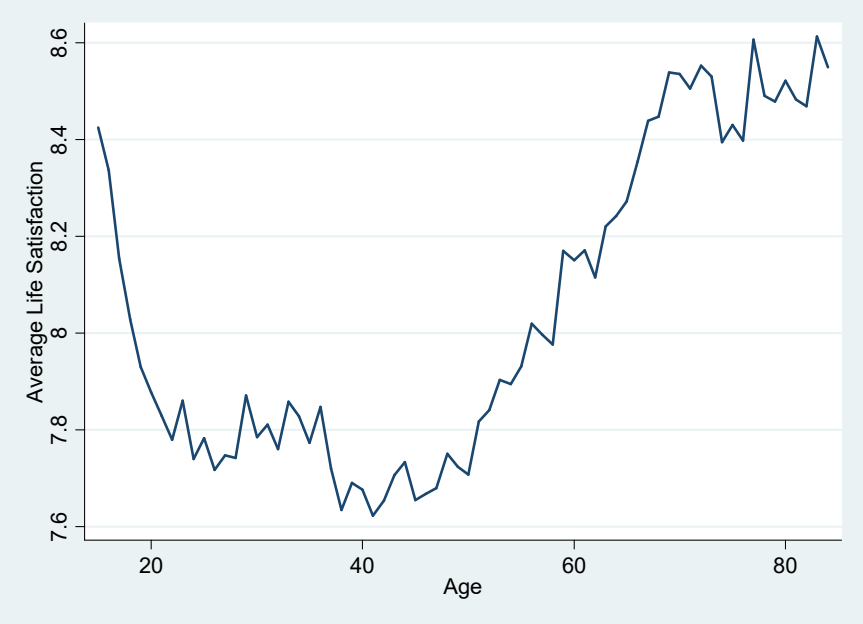
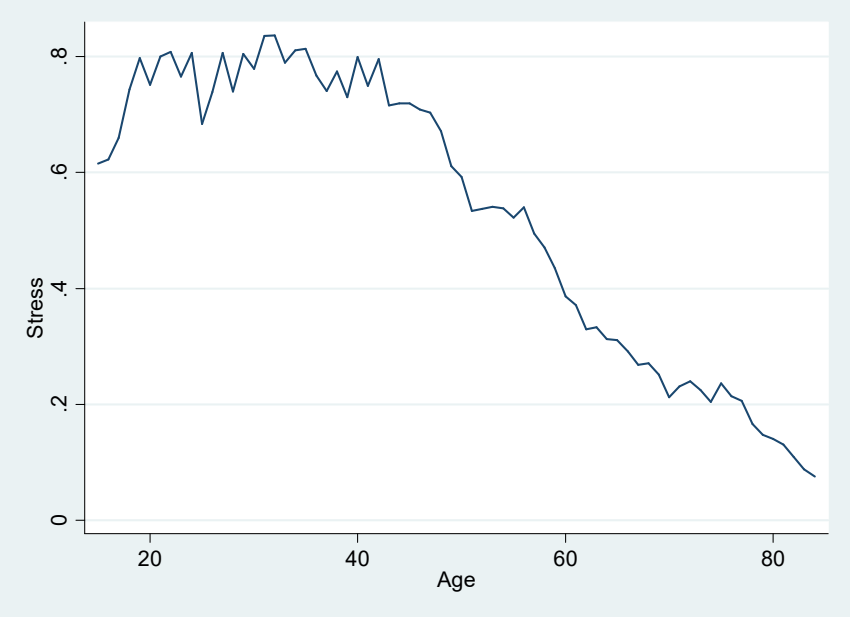
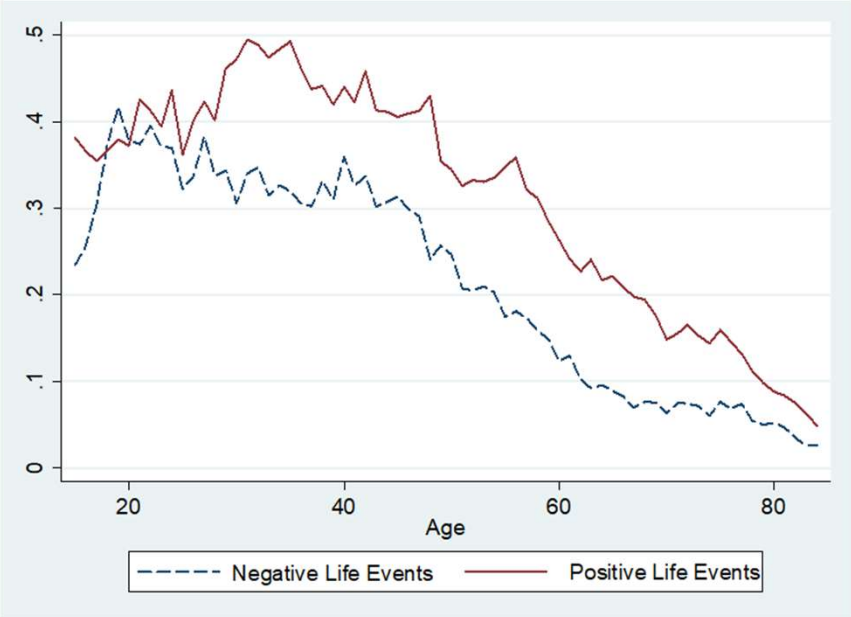


Figure 1b: Average stress level for Australians aged 15 to 84

Task: What causes this U-shape in happiness over time?







$$GS_t = C + \delta_1(Positive\ _Stress_t) + \delta_2(Negative\ _Stress_t) + \varepsilon_t$$

-1.76 (5.13)

-0.57 (1.58)

$R^2 = .82$

What about the effect of Personality on Life Satisfaction

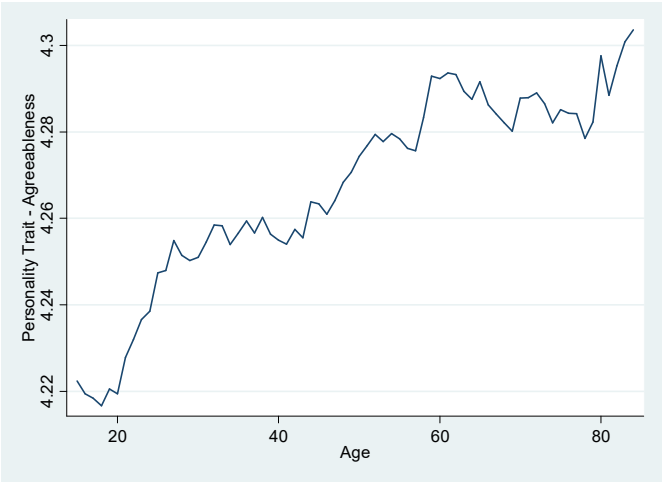


Figure 3a: Agreeableness by Age

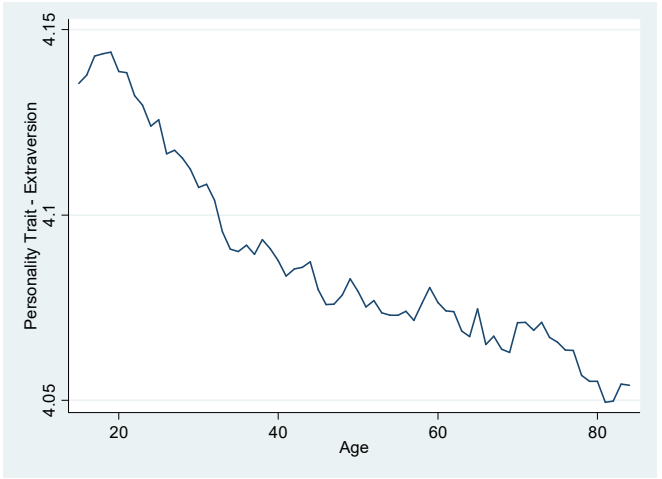


Figure 3b: Extraversion by Age

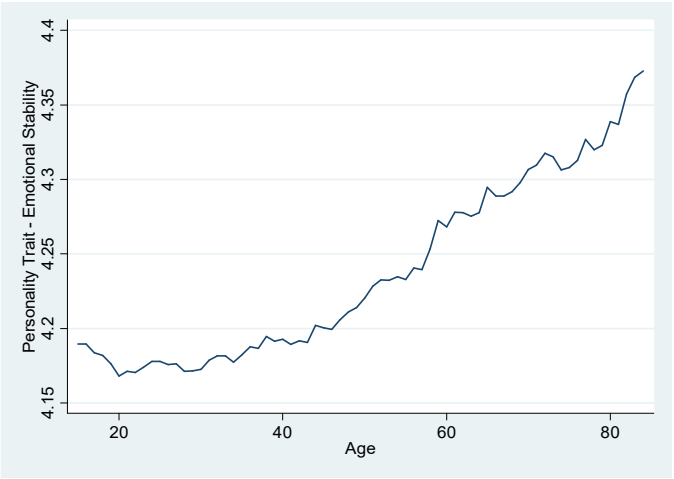


Figure 3c: Emotional Stability by Age

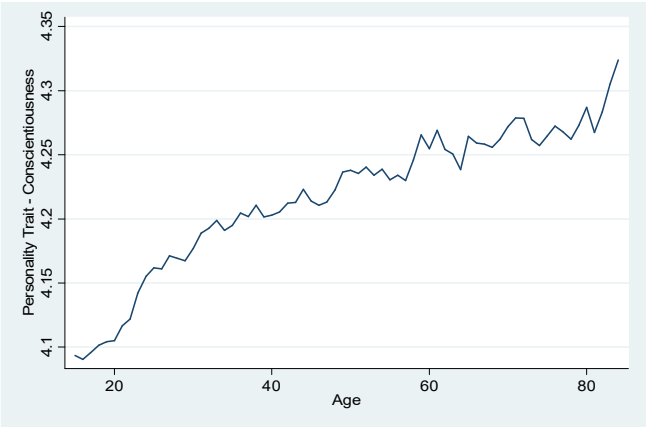


Figure 3d: Conscientiousness by Age

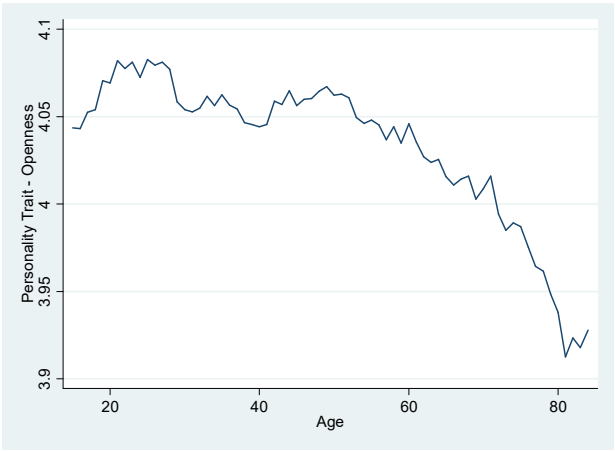


Figure 3e: Openness by Age

What about the effect of Personality on Life Satisfaction

$$GS_t = C + \delta(Stress_t) + \lambda_1 Pa_t + \lambda_2 Pex_t + \lambda_3 Pem_t + \lambda_4 Pc_t + \lambda_5 Po_t + \varepsilon_t$$

9.35	-1.54	-0.21	3.84	-0.74	-1.40	-1.53
(2.54)	(10.91)	(0.29)	(5.62)	(1.03)	(2.22)	(2.43)

$R^2 = .88$

Life Satisfaction and Stress and Personality

	Stress (2)		Stress Valency (3)		Stress + Direct Personality (4)		Stress + Direct Personality + Indirect Stress*Personality (5)	
Variable:	**coefficient**	**t-value**	**coefficient**	**t-value**	**coefficient**	**t-value**	**coefficient**	**t-value**
Stress/1000	-1.18	(17.21)			-1.54	(10.91)	-0.68	*(2.54)*
Positive_Stress/1000			-1.76	(5.15)				
Negative_Stress/1000			-.572	(1.58)				
Average Personality								
Agreeableness					-0.21	(0.29)	-0.55	(1.07)
Extraversion					3.84	(5.62)	1.62	(2.64)
Emotional Stability					-.074	(1.03)	-0.17	(0.34)
Conscientiousness					-1.40	(2.22)	0.004	(0.01)
Openness/Intellect					-1.53	(2.43)	-0.39	(0.73)
Stress * Personality								
Agreeableness * Stress							4.15	(0.95)
Extraversion* Stress							10.43	(5.71)
Emotional Stability * Stress							0.44	(0.22)
Conscientiousness * Stress							1.83	(0.79)
Openness/Intellect * Stress							-8.65	(5.17)
constant	8.67	(215.43)	8.73	(177.83)	9.35	(2.54)	6.63	(2.42)
R^2	0.81		0.82		0.889		0.95	
Adjusted R^2			0.815		0.879		0.945	

Panel: leads and lags

Because panel data contain information on change over time, we can analyse the change in phenomena from a period to others: Examples:

Panel: leads and lags

Because panel data contain information on change over time, we can analyse the change in phenomena from a period to others: Examples:

$$LS_{it} = \beta_1 + \beta_2 MH_{it-1} + u_i + v_{it}$$; t-1 lag (1)

Stata: gen mhL1= mental_health[_N-1]
xtreg lifesat mhL1 , fe cluster(persnr)

$$LS_{it} = \beta_1 + \beta_2 MH_{it-2} + u_i + v_{it}$$; t-2 lag (2)

Stata: gen mhL2= mental_health[_N-2]
xtreg lifesat mhL2 , fe cluster(persnr)

$$LS_{it} = \beta_1 + \beta_2 MH_{it+1} + \varepsilon_{it}$$; t+1 lead (3)

Stata: gen incomeLd1= lnhousehold_income [_N+1]
xtreg lifesat incomeLd1, fe cluster(persnr)

$$LS_{it+1} = \beta_1 + \beta_2 MH_{it} + \varepsilon_{it}$$; t+1 lead (4)

Stata: gen lifesatlead1= lifesat[_N+1]
xtreg lifesatlead1 mental_health, fe cluster(persnr)

Task: 1. Does panel analysis provide causal estimates?
2. Can leads or lags be used as an instrument?

Panel lead and lag Results

	(1)	(2)	(3)
	Life Satisfaction	**Life Satisfaction**	**Life Satisfaction**
age	0.0027**	-0.0012	-0.0018*
	(0.0007)	(0.0008)	(0.0008)
Ln Household Income	0.0303**	0.0346**	0.0366**
	(0.0039)	(0.0042)	(0.0043)
female	0.0000	0.0000	0.0000
	(.)	(.)	(.)
educ years	-0.0163**	-0.0120**	-0.0097**
	(0.0024)	(0.0025)	(0.0026)
bachelor	0.0327+	0.0135	0.0070
	(0.0173)	(0.0184)	(0.0187)
Mental Health	0.0258**		
	(0.0003)		
Mental Health t-1		0.0075**	
		(0.0002)	
Mental Health t-2			0.0030**
			(0.0002)
N	230857	230857	230857
R-squared	0.0911	0.0095	0.0023

Marginal effects; Standard errors in parentheses
+ p<0.10, * p<0.05, ** p<0.01

XT: Binary Choice & Multinomial Models in Stata

Time series analysis in Stata can be done for many of the analysis methods we have discussed: Eg:

help xtprobit:

Random-effects (RE) model:

xtprobit depvar [indepvars] [if] [in] [weight] [, re RE_options]

Population-averaged (PA) model

xtprobit depvar [indepvars] [if] [in] [weight] , pa [PA_options]

help xtlogit

Random-effects (RE) model

xtlogit depvar [indepvars] [if] [in] [weight] [, re RE_options]

Conditional fixed-effects (FE) model (i.e. Fixed effect Logit)

xtlogit depvar [indepvars] [if] [in] [weight] , fe [FE_options]

Population-averaged (PA) model

xtlogit depvar [indepvars] [if] [in] [weight] , pa [PA_options]

XT: Models in Stata

help xtivreg

GLS random-effects (RE) model

xtivreg depvar [varlist_1] (varlist_2 = varlist_iv) [if] [in] [, re RE_options]

Between-effects (BE) model

xtivreg depvar [varlist_1] (varlist_2 = varlist_iv) [if] [in] , be [BE_options]

Fixed-effects (FE) model

xtivreg depvar [varlist_1] (varlist_2 = varlist_iv) [if] [in] , fe [FE_options]

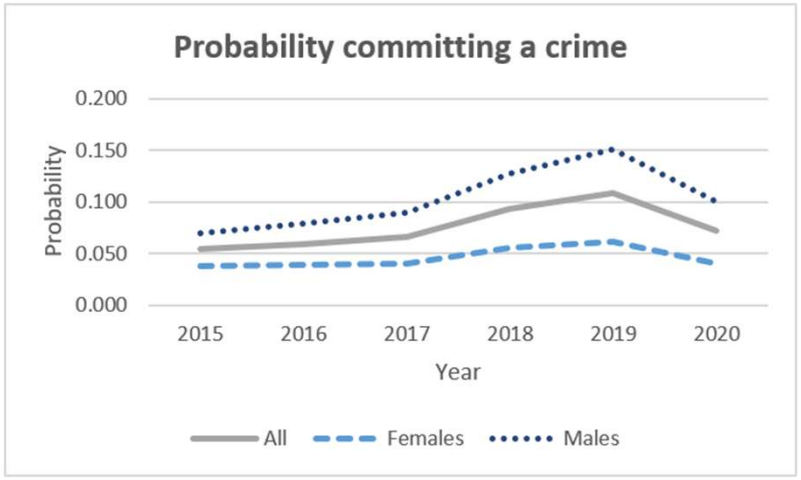
First-differenced (FD) estimator

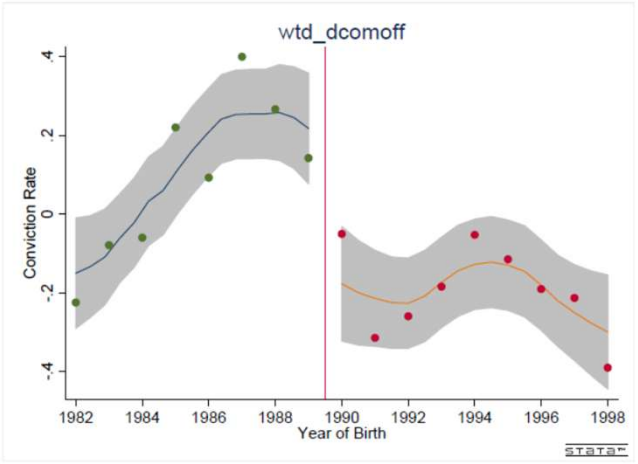
xtivreg depvar [varlist_1] (varlist_2 = varlist_iv) [if] [in] , fd [FD_options]

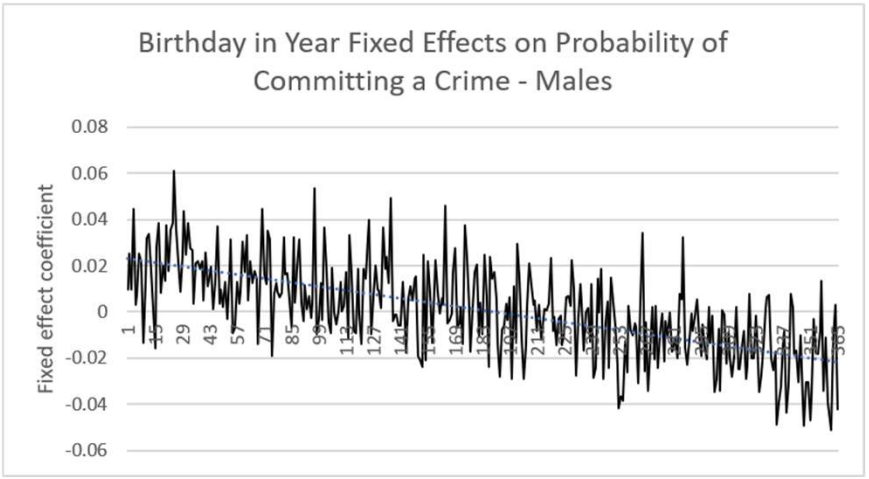
Using Time Series Data for Natural Experiments

Calendar Year	2000	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
QED						Year 7 moved to High School					
Administration data available											
NAPLAN data available											
KLA data available											
First Prep Year Level		4	5	6	7	8	9	10	11	12	
Age First Prep Year		9	10	11	12	13	14	15	16	17	18
Age			10	11	12	13	14	15	16	17	18
Student Birth Year	Enrolment	9									
1 Jan - 31 Dec 1994	1	12	12	12	12	12	12	12	12	12	12
Jan - Dec 1995	2	11	12	12	12	12	12	12	12	12	12
Jan - Dec 1996	3	10	11	12	12	12	12	12	12	12	12
Jan - Dec 1997	4	9	10	11	12	12	12	12	12	12	12
Jan - Dec 1998	5	8	9	10	11	12	12	12	12	12	12
Jan - Dec 1999	6	7	8	9	10	11	12	12	12	12	12
Jan - Dec 2000	**7**	**6**	**7**	**8**	**9**	**10**	**11**	***12***	***12***	***12***	***12***
Jan - Dec 2001	**8**	**5**	**6**	**7**	**8**	**9**	**10**	***11***	***12***	***12***	***12***
1 Jan 2002 -30 Jun 2002 (ITT)	9	5	6	7	8	9	10	***11***	***12***	***13***	***13***
Jul 2002 - Jun 2003	10		5	6	7	8	9	***10***	***11***	***12***	***13***
Jul 2003 - Jun 2004	11			5	6	7	8	9	10	11	12
Jul 2004 - Jun 2005	12				5	6	7	8	9	10	11
Jul 2005 - Jun 2006	13					5	6	7	8	9	10
Jul 2006 - Jun 2007	14						5	6	7	8	9
Jul 2007 - Jun 2008	15							5	6	7	8
Jul 2008 - Jun 2009	16								5	6	7
Jul 2009 - Jun 2010	17									5	6

Time allows us to look at data from different perspectives







Using Time Series Data for Natural Experiments

	P(Committing an Offence)			Number of Offences		
	(4)	(5)	(6)	(7)	(8)	(9)
	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education						
Preparatory Year of Education Treatment	0.841**	0.838**	0.843**	0.841**	0.838**	0.843**
	(0.002)	(0.004)	(0.004)	(0.002)	(0.004)	(0.004)
F-stat. [3]	1.2e+05	59156.29	57648.72	1.2e+05	59156.29	57648.72
Stage 2						
Years of Education	-0.0715**	**-0.0983****	**-0.0345****	-0.1114**	**-0.1557****	**-0.0425****
	(0.0017)	(0.0027)	(0.002)	(0.0101)	(0.0171)	(0.0087)
Fixed Effects						
Year	yes	yes	yes	yes	yes	yes
Birth-day-in-year	yes	yes	yes	yes	yes	yes
Number of Individuals	176,929	93694	83,235	176,929	93694	83,235
N	489466	255373	234,093	489466	255373	234,093
Under identification test[1]	1.2e+04	2.5e+04	2.2e+04	1.2e+04	2.5e+04	2.2e+04
Weak identification test[2]	4.7e+05	5.90e+04	5.8e+04	4.7e+05	5.9e+04	5.8e+04

Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01

[1] Kleibergen-Paap rk LM statistic; [2] Kleibergen-Paap rk Wald F statistic

[3] Exceeds F-stat requirements per Lee et. al., 2021

End of Session